

Methods of the Conley index for a class of functionals in Banach spaces involving p-Laplacian

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We will be concerned with a class of functionals involving the p-Laplacian. The natural setting for these functionals is the Banach space $X = W_0^{1,2p}(\Omega)$, where $p > n/2$ and $\Omega \subset \mathbb{R}^n$ is a bounded domain with sufficiently smooth boundary.

To a given functional $f: X \rightarrow \mathbb{R}$, we associate a quasilinear elliptic problem (EL), defined by the corresponding Euler-Lagrange equation. The critical points of f are precisely the solutions of (EL). Since X is not isomorphic to its dual space, the standard notion of nondegeneracy of critical points is not applicable in this setting. Moreover, f serves as a Lyapunov function for a vector field which, in our setting, is merely continuous and therefore need not generate a flow.

We develop a method based on an extension of the Conley index to Banach spaces for merely continuous vector fields. This framework allows us to overcome the difficulties arising from the lack of a flow and the absence of the usual nondegeneracy condition, and to establish existence and multiplicity results for solutions of (EL) in the presence of $\mathbb{Z}_2$ -symmetry.